Testing the Built-in Microphone (PDM)

Objective:

To understand the working of the built-in PDM microphone on the Arduino Nano 33 BLE Sense Lite and observe audio sample values generated from sound.

(Note: Install PDM Library **)**

Theory:

The Arduino Nano 33 BLE Sense Lite contains a built-in digital microphone. The microphone captures sound waves from the environment and converts them into digital audio samples that can be processed by the microcontroller.

PDM stands for Pulse Density Modulation. It is a digital method used by the microphone to transmit audio data to the nRF52840 microcontroller.

The microphone does not directly recognize words. Instead, it converts sound into numerical values called audio samples.

Example: Voice → Microphone → Audio Samples → CPU

Procedure:

1. Connected the Arduino Nano 33 BLE Sense Lite to the laptop.

2. Opened Arduino IDE.

3. Navigated to:File → Examples → PDM → PDMSerialPlotter.

4. Uploaded the example program to the board.

5. Opened the Serial Monitor/Serial Plotter.

6. Observed the continuously changing microphone values.

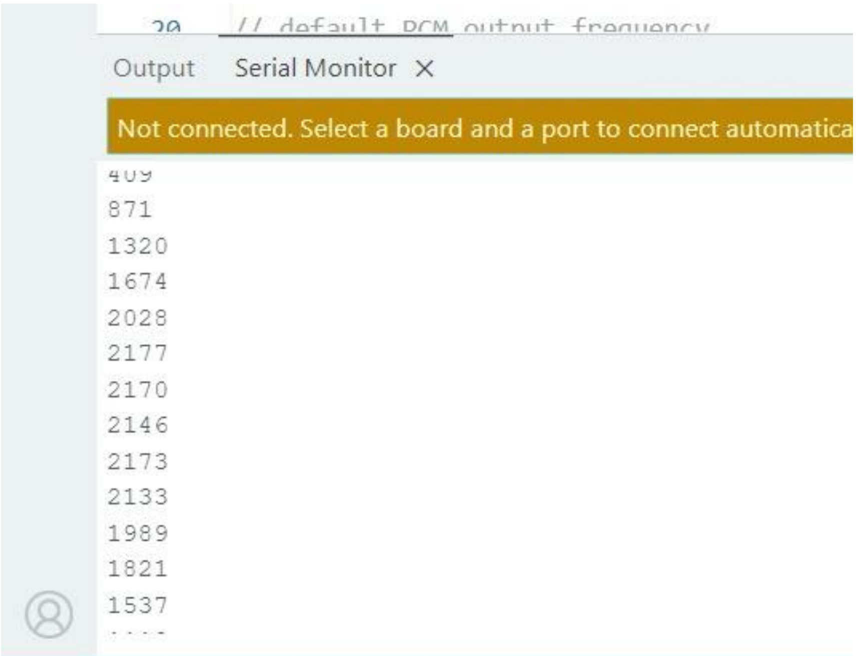
7. Spoke near the microphone and clapped hands to observe changes in the values.

Observation:

During silence, the values remained relatively small.When speaking or clapping, larger variations in the values were observed.These values represent raw audio samples and are not sound levels in decibels (dB).

Without Sound value:

With Sound value:



Working Principle:

Sound waves in air create vibrations.

Air Vibrations

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Microphone

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Audio Samples

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nRF52840 CPU